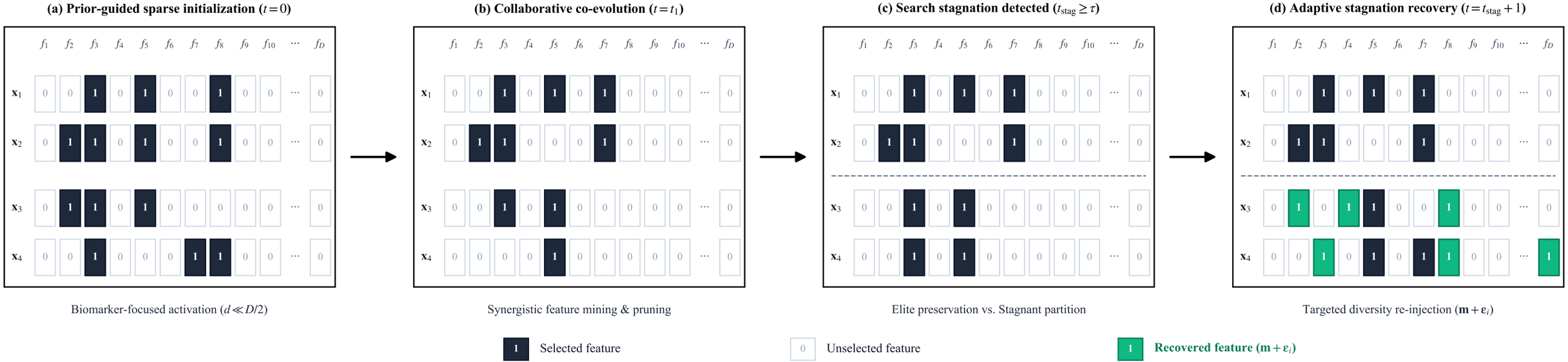
(a) Prior-guided sparse initialization ($t=0$)
(b) Collaborative co-evolution ($t=t_1$)
(c) Search stagnation detected ($t_{\text{stag}} \geq \tau$)
(d) Adaptive stagnation recovery ($t=t_{\text{stag}}+1$)
f_1 f_2 f_3 f_4 f_5 f_6 f_7 f_8 f_9 f_{10} ⋯ f_D
$\mathbf{x}_1$
$\mathbf{x}_2$
$\mathbf{x}_3$
$\mathbf{x}_4$
Biomarker-focused activation ($d \ll D/2$)
Synergistic feature mining & pruning
Elite preservation vs. Stagnant partition
Targeted diversity re-injection ($\mathbf{m}+\boldsymbol{\varepsilon}_i$)
Selected feature
Unselected feature
Recovered feature ($\mathbf{m}+\boldsymbol{\varepsilon}_i$)